MODULE *MixedIsolation*

EXTENDS *Naturals*, *Sequences*, *FiniteSets*

CONSTANTS *Keys*, *Values*, *InitVal*, *Workload*

$TxnIds \triangleq \text{DOMAIN } Workload$
$InitTx \triangleq 0$
$AllTxns \triangleq TxnIds \cup \{InitTx\}$

$IsoLevels \triangleq \{\text{"RA"}, \text{"CC"}, \text{"PC"}, \text{"PSI"}, \text{"SI"}, \text{"SER"}\}$

VARIABLES *client_dep*,
client_last_commit,
router_ct,
shard_ct,
tx_status,
tx_snapshot,
tx_checkset,
tx_buffer,
tx_commit_time,
pc,
store,
op_seq

all_vars $\triangleq$
⟨*client_dep*,
client_last_commit,
router_ct,
shard_ct,
tx_status,
tx_snapshot,
tx_checkset,
tx_buffer,
tx_commit_time,
pc,
store,
op_seq
⟩

1. OPERATIONAL PROTOCOL

$Init \triangleq$
LET $Sessions \triangleq \{Workload[t].session : t \in TxnIds\}$
IN $\land\ client\_dep = [s \in Sessions \mapsto 0]$
$\land\ client\_last\_commit = [s \in Sessions \mapsto 0]$